Day	Main focus
Friday	Cloud fundamentals, virtualization, AWS infrastructure, development tools and IAM
Saturday	Compute, databases, storage and caching
Sunday	Serverless, message queues, comparisons and timed case studies

Lesson 1 Cloud Computing Fundamentals

What is Cloud Computing

Cloud Computing is the on-demand delivery of IT resources through the internet by pay-as-you-go pricing.

Three Simple Model

IaaS – Infrastructure as a Service
The provider supplies an infrastructure such as virtual machines, networking and storage

Example: **Amazon EC2**

Question Clues:

"Need full control", "Must be able to configure or choose their own operating system", "Similar to traditional infrastructure", "Need virtual machine".

PaaS – Platform as a Service
The provider manages an infrastructure, operating system and application platform. The customer mainly deploys and manages the application and data.

Example: **AWS Elastic Beanstalk**

Question Clues:

"Deploy the application quickly", "Does not want to manage the servers or operating system", "Wants to focus on application development"

SaaS – Software as a Service
The provider supplies a complete application for users to access and use.

Example: **Gmail, Microsoft 365**

Provider manages almost everything and customers only use or configure the software

Question Clues:

"Can be accessible through browser", "End-user application", "Complete Software", "No installation or infrastructure management".

Conclusion:

IaaS: Rent the machine	**PaaS**: Deploy the application for you	**SaaS**: Use the finished application

Control decreases in this order: On-premise → IaaS → PaaS → SaaS

The Four Deployment Models

Describes **Where and for whom** the cloud environment operates

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Public Cloud

Infrastructure that operates by the cloud provider and is offered to customers through the internet. It is highlight scalable, usually cheaper, no in-house physical hardware infrastructure required, and less direct control over the underlying infrastructure.

Example: Using AWS infrastructure for an online store

Private Cloud

Cloud infrastructure that's dedicated to one organization

Greater control and privacy, Suitable for strict organization rules, More expensive, Usually requires more internal expertise

Example: A government agency operating an environment exclusively for its own departments

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Hybrid Cloud

Combines cloud resources with private or on-premises infrastructure.

Example: A bank keeps sensitive data in it's own private infrastructure but runs it's public website in AWS.

Community Cloud

Infrastructure that's shared by multiple organizations with similar requirements

Example: Several healthcare organizations sharing infrastructure designed around common healthcare regulations.

Memory Shortcut:
Public: open provider service
Private: one organization
Hybrid: mix of cloud and on-premise/private
Community: shared infrastructure with similar organisations

Lesson 2 Virtualization and AWS infrastructure

Virtualization is a way to separate a single strong computer into multiple separate computers. A technology that abstracts physical hardware resources to create virtual versions of servers, operating systems, storage and networks

Benefits: Cheaper cost, easier to maintain (centralized maintenance and management), more scalable and flexible, easier backup and disaster recovery, better hardware utilisation, faster creation of servers.

What is a virtual machine:
A virtual machine is a software-based representation of a physical computer that can run its own operating system and applications.

What is hypervisor: There's two types of hypervisor

A hypervisor is software that creates and runs, and manages virtual machines while allocating physical resources to them.

Sit between the physical hardware and virtual machines / Between a host operating system and virtual machines

Type 1: Bare-metal Hypervisor

Runs **directly on the physical hardware**

It does not require a normal host operating system underneath, Better performance and efficiency, Commonly used in data centres

Examples: VMware ESXi and Microsoft Hyper-V

Type 2: Hosted Hypervisor

Runs **on top of an existing operating system**

It is easier for an ordinary user to install, common for development testing and personal use, has additional cost from the host operating system

Examples: Oracle VirtualBox and VMware Workstation

Hardware → Host Operating system → Hypervisor → Virtual machines

Type 1 touches the hardware; Type 2 sits on an OS

Five types of virtualization

Server virtualization
Divides one physical server into multiple virtual machines

Network Virtualization
Creates logical software-based network from physical networking resources

Application Virtualization
Delivers an individual application without relying on a traditional local installation

Desktop Virtualization

Separates the entire desktop environment from the user's physical device

Storage Virtualization

Combines multiple physical storage devices into one logical storage pool

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AWS Global Infrastructure

AWS Region

An AWS Region is a separate geographical area containing multiple Availability Zones.

Example: Resources are not automatically replicated from one region to another. The customer decides whether it is necessary for cross-Region replication

Four factors for selecting a Region.

(1) Data governance and legal requirements, (2) Proximity to users and latency, (3) Availability of required services, (4) Cost

The organization should select a Region close to its users to reduce latency, provided that it meets the governance requirements and has availability. Meets its legal, data-governance and service requirements

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Availability Zone

Availability zone is an isolated location of an AWS Region, consisting of one or more data centres.

More availability zone, less chance for website to go down!

Date Centre

A data centre is physical facility containing servers, storage, networking equipments, power and cooling systems.

AWS Region → Availability Zone → Data Centre

Edge Location

An edge location is a location closer to end users that caches and delivers content with lower latency.

Amazon CloudFront uses edge locations as part of its **Content Delivery Network**, or CDN

Example:
A Malaysian user requests an image from an American server, instead of retrieving that image from the American server every time, CloudFront may use a cached data to send over that image from a nearby edge location.

Lesson 3 Development, AWS Services and IAM

Cloud Development Life Cycle

SDLC: Systems Development Life Cycle:
Plan → Define → Design → Develop → Deploy → Maintain

Plan:
Determine the project's purpose, feasibility, resources, time and cost

Define:
Gather and Document System Requirements

Design:
Plan the architecture, database, interfaces and system components.

Develop:
Write the code, install components and create the system.

Maintain:
Monitor the system, correct problems and implement improvements

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Waterfall versus Agile

Waterfall is sequential SDLC where each phase of development is completed before the next phase begins.

Agile is an iterative methodology where development is in which software is delivered in short cycles called sprints, allowing frequent delivery, feedback and improvement. Commonly lasting one to four weeks

Why Cloud Computing supports Agile:

Cloud resources can be provisioned without waiting to purchase hardware. Developers can create temporary environments, test an application, receive feedback and change the resources as requirements evolve.

Development Activities

Develop
Write and review source code

Build
Compile code and perform unit tests

Test
Integration, load, interface and penetration testing

Deploy
Release code to production

Maintain
Monitor the production system and detect errors

AWS Service Recognition

Compute

Amazon EC2
Resizable virtual machines

EC2 Auto Scaling
Automatically adds or removes EC2 instances

Elastic Beanstalk
Deploy and scales web applications

AWS Lambda
Runs code without provisioning or managing servers

Storage

Amazon S3
Object Storage

Amazon EBS
Block Storage primarily attached to EC2

Amazon EFS
Shared File storage

S3 Glacier storage classes
Low-cost archival storage

Database

Amazon RDS
Managed relational databases

Amazon Aurora
AWS relational databases compatible with MySQL/PostgreSQL

Amazon DynamoDB
NoSQL key-value and document database

Amazon Redshift
Data warehouse and large-scale analytic

Networking and content delivery

Amazon VPC
Logically isolated virtual network

Elastic Load Balancing
Distributes traffic across multiple targets

Amazon CloudFront
Content Delivery Network

Amazon Route 53
Domain Name System service

AWS Direct Connect
Dedicated private connection to AWS

AWS VPN
Encrypted private network tunnel

Security

AWS is responsible for security of the cloud
The customer is responsible for security in the cloud

Browser based IDE: Cloud9

Method	**Meaning**
AWS Management Console	Graphical interface operated through a browser
AWS CLI	Text commands entered through a terminal
AWS SDK	Language-specific libraries used inside application code
Service API	Direct programmatic requests to an AWS service

Four IAM Components

IAM user

A specific person or application with an identity in the AWS account. It may have longer-term credentials.

IAM group

A collection of IAM users that could receive similar permissions.

IAM policy

A document defining permissions

IAM Policy Types:

Identity-based policy

Attached to a user, group or role (e.g. "What can this identity do")

Resource-based policy

Attached directly to a resource, such as an S3 bucket (e.g. "Who can access this resource")

IAM role

An assumable identity containing permissions

Lesson 4 AWS EC2 and Elastic Beanstalk

Compute

Compute refers to the required resources to run programs, including CPU, memory, networking and related resources

AWS provides **different compute services** because applications requires different **level of control, management and scalability.**

Four major compute approaches

Virtual machine/IaaS: EC2

Customer manages and controls virtual servers

EC2 Definition:
Amazon Elastic Compute Cloud provides secure, resizable virtual machines called EC2 instances in the AWS Cloud.

PaaS: Elastic Beanstalk

Customer uploads application code

Serverless: Lambda

Runs code without managing servers

Containers: ECS, SKS, Fargate

Runs containerized applications

What is an AMI

An Amazon Machine Image is a template used to launch EC2 instances.

An AMI normally means: Operating system, initial software, configuration and storage mapping.

AMI = template. EC2 instance = running virtual machine created from that template.

What is an instance type

An EC2 instance type defines the memory, CPU, storage and networking capacity provided to an instance.

5 categories of instance types:

*General Purpose (**Balanced Application or ordinary web server**), Compute Opimised (**Video-Editing or Scientific Calculations**), Memory Optimised (**Large in-memory***

database)**, Accelerated Computing (**GPU or Machine Learning heavy)**, Storage Optimised (**Very high local storage performance)

EC2 launch decisions:
AMI, Instance Type, Network, IAM role, User Data, Storage, Tags, Security Group, Key Pair

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Security Group

A security group is a virtual firewall that controls traffic to and from AWS resources such as EC2 instances.

Example: *Permit SSH traffic on TCP port 22 only from the administrator's IP address*

Key Pair
Public Key: Store by AWS/ on the instance
Private Key: Kept securely by the customer

For Linux, the private key is usually used for SSH access, Never expose or share private key.

IAM role on EC2 instance
If an EC2 application needs to access S3,
Do not store an IAM user's permanent access key on the server
Attach an IAM role to the EC2 instance
The application receives temporary credentials through that role

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User Data

EC2 user data is an optional script that runs when an instance is first launched.

Like: Software installation, Web-server configuration, Downloading Application Files

This is an example of automation

EC2 Lifecycle

Pending→Running→Stopping→Stopped→Running

Or:

Running → Shutting Down → Terminated

Elastic IP address
IP address that is public IPv4 address and can be associated with an EC2 instance. Should not change even after stop/start

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Amazon CloudWatch
Amazon Cloudwatch monitors AWS resources and applications by collecting metrics.

For EC2:
CPU, Networking, Instance health

*** CloudWatch can help determine when scaling is required**

EC2 Auto Scaling

EC2 Auto Scaling automatically adds or removes EC2 instances according to defined conditions

Example: CPU exceeds 70% usage, launch additional instances, traffic decreases, remove unnecessary instances.

Elastic Load Balancing

Elastic Load Balancing distributes incoming traffic to multiple targets, such as EC2 instances. It prevents one server from handling all requests.

Auto Scaling and Load Balancing commonly work together:
Load balancer distribute incoming traffic, Auto Scaling adds instances when demand rises, Load balancer then begins routing traffic to the new healthy instances, Auto Scaling removes unnecessary instances when demand falls.

AWS Elastic Beanstalk

AWS Elastic Beanstalk is a PaaS service used to deploy, scale and manage web applications.

Amazon EC2	Elastic Beanstalk
IaaS	PaaS
Customer manages the operating system	Platform handles more infrastructure management
Greater control	Easier deployment
Manually configure more components	Automatically provisions an application environment
Suitable for custom server requirements	Suitable for ordinary web applications

AWS Elastic Beanstalk is suitable because it allows the developers to upload their Node.js application code while automatically handling infrastructure provisioning, deployment, load balancing, automatic scaling and health monitoring. This reduces management work, improves developer productivity and allows the application to scale according to traffic.

If the case instead says the company needs complete operating-system control and specialised server configuration, choose **Amazon EC2**.

Lesson 5 Relational Databases, Amazon RDS and DynamoDB

What is a database

A database is an organized collection of data that can be stored, retrieved and managed electronically.

A database management system or a DBMS is a software used to manage that data.

Relational databases

A relational database stores structured data in tables consisting of rows and columns with relationships established between tables

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Relational vs NoSQL database

Relational database has data commonly organized into tables, NoSQL database can use key-value, document and other models, it also avoids JOIN operations unlike relational database, it is more flexible data structure, queries not using SQL but using Service-specific APIs or query language, it is suitable for flexible data and large-scale access patterns unlike relational where it is suitable for strongly related structural data. DynamoDB is a NoSQL database and MySQL and PostgreSQL is an example of relational database.

Amazon RDS Definition

Amazon Relational Database Service, or Amazon RDS is managed AWS service used to set up, operate and scale relational database

Choose RDS when a company wants to manage a relational database and reduce administrative work.

RDS Automated backup and snapshots

Automated backup

Automated backup is managed according to the configured backup policy, it also supports point-in-time recovery within the retention period, it is useful for recovering accidental changes.

Manual DB snapshot

Created manually by customer, retained until it's deleted, represents the complete DB instance storage, Restoration creates a new DB

RDS Multi A-Z deployment

A Mult-AZ deployment replicates the database to another Availability Zone to provide high availability and failover support

Multi-AZ survives failures. Read replica handles readers.

RDS = managed database service. Aurora = AWS relation database engine

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DynamoDB

a fully managed serverless NoSQL database service supporting key-value and document data models.

Lesson 6: Amazon S3, EBS, EFS and Archival Storage

Three main storage types

Object Storage

Object stored inside buckets **[Amazon S3]**

Block Storage

Storage volume divided into blocks **[Amazon EBS]**

File Storage

Files organized into folders/directories **[Amazon EFS]**

S3 stores objects, EBS acts like a disk, EFS acts like a shared folder

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Amazon S3

Amazon Simple Storage Service, or Amazon S3, is a scalable object-storage service that stores data as objects inside buckets.
S3 is suitable for Images and Videos, Backups, Log files, Documents

S3 is not normally used as an EC2 operating system disk, that is EBS
Traditional shared file system is EFS
Relational database is RDS

S3 Components

Bucket

A bucket is created in an AWS Region and it is a container to store S3 objects.

Object

An object is the actual data stored in S3, together with its metadata.

Object key

An object key is unique name or path used to identify an object within a bucket

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S3 Storage Classes

Storage classes allow the customer to balance storage cost, access frequency, retrieval speed, availability requirements.

S3 Versioning

S3 Versioning stores multiple versions of an object, allowing retrieval of previous versions after accidental deletion or overwriting.

Amazon EBS

Amazon EBS, Amazon Elastic Block Store provides persistent block-storage volumes that can be attached to EC2 instances.

EBS and Availability Zones

An EBS volume and the EC2 instance to which it is attached must normally be in the same Availability Zone

EBS Snapshot

EBS Snapshot is a point-in-time backup of an EBS volume

Amazon EFS Definition

Amazon Elastic File System provide elastic, shared file storage that can be accessed concurrently by multiple compute resources.

EFS is suitable when multiple EC2 instances require access to the same file.

One server needs a disk = EBS. Multiple servers need shared file = EFS

Lesson 7: CloudFront, ElastiCache and Caching Strategies

What is caching
Caching stores frequently or recently requested data in a faster temporary location so future requests an served more quickly.

Without caching:
User → Application → Database/login → Response

With caching:
User → Application → Cache → Faster Response

Caching can reduce response time, database workload, it can reduce origin-server traffic, improve scalability, improve user experience, potentially reduce processing costs. The original database or original server usually remains the source of truth.

Cache hit and cache miss

Cache hit
A cache hit occurs when the requested data is available and valid in the cache. The data is returned without querying the original source.

Cache miss
A cache miss occurs when the requested data is not available or expired. The application tries to receive the cache / requested data but is not available or has expired.

The application will then retrieve the original source and may store it in cache for future requests.

Hit = found in cache, miss = retrieve from source

Time to Live

Time to Live, or TTL specifies how long cached data remains valid before it expires.
Product information TTL: 10 minutes
News homepage TTL: 1 minute
Static logo TTL: several days

TTL trade-off
A short TTL has fresher data while long TTL has a higher cache-hit rate, a short TTL will have more source requests, while a long TTLE will have fewer source requests. A short TTL has lower risk of stale data, and a long TTL has a greater risk of stale data.

Stale data is an outdated cached value that no longer matches the original source

If a product changes its price from RM50 to RM40, and the cached value is still RM50, the cached value is stale.

Cache invalidation and eviction

Cache invalidation

Cache invalidation deliberately removes or updates cached content because the original data has changed.

Cache eviction

Cache eviction removes items automatically, commonly because the cache is full or an expiration policy applies

Invalidation is content that is purposely removed because of change, and Eviction is update of cache because expire policy has passed or cache is full.

Amazon CloudFront

Amazon CloudFront is a Content Delivery Network that delivers static and dynamic web content through edge locations to users.

It reduces latency by serving cached content from an edge location near the user

CloudFront Components

Origin

The origin is the original location where CloudFront receives content

Edge Location

An edge location is a CloudFront location near users that caches and delivers content.

Distribution

A CloudFront distribution defines how CloudFront retrieves, caches and delivers content from an origin.

So basically, Origin is the original content, Edge location is the nearest location to the user for CloudFront to cache the content, and Distribution is how CloudFront caches the content, delivery configuration

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Amazon ElastiCache

Amazon ElastiCache is a managed in-memory caching system to improve a application and database performance

In-memory means the cached data is stored mainly in the RAM, which is faster to access than data on ordinary storage.

Caching Strategies

Cache-aside or lazy loading

Cache-aside retrieves data from the database only after a cache miss and then stores the result in the cache

Write-through caching

Write-through caching updates the cache when the primary database is updated

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Combining strategies with TTL

Cache-aside and write-through can be combined with TTL because write-through keeps actively updated values content, while cache-aside handles missing or expired values. TTL eventually removes old values.

DynamoDB Accelerator

DAX definition

DynamoDB Accelerator, or DAX, is a managed in-memory cache designed specifically for DynamoDB.

Website content near users = CloudFront

Database Results in memory = ElastiCache

DynamoDB-specific cache = DAX

Lesson 8: AWS Lambda, API Gateway, SQS and SNS

What does serverless mean

Serverless computing allows developers to run applications without provisioning or managing the underlying servers.

Advantages of serverless computing

It doesn't need server provisioning, it has automatic scaling, it reduces infrastructure administration, pays according to usage, It has faster application deployment, it is suitable for event-driven application, and resources can be scale down when unused.

Serverless limitations

It has less infrastructure control, the execution-duration and resource limits, it possible cold-start delay, applications should normally be stateless, Monitoring distributed functions can be complex, Not every long-running or constantly active workload is suitable.

AWS Lambda

Lambda definition

AWS lambda is a serverless compute service that runs code in response to events without requiring the customer to provision or manage servers

Lambda Function

A lambda function is a block of application code that performs a specific task when invoked.

Stateless Function

A stateless function does not depend on locally stored information from a previous invocation

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Amazon API Gateway

Amazon API Gateway is a managed service used to create, publish, secure and monitor APIs. API Gateway can provide an HTTP endpoint which a web or mobile application invokes Lambda.

Example: First, the user submits an order through a website, then the website send an API request, then the API Gateway receives the request and invokes Lambda next. Lambda then validates and process the order, and the Lambda stores the order in DynamoDB, which returns a response through API Gateway.

API Gateway Receives API, Lambda Process Logic Code

Decoupling

Decoupling is: Decoupling reduces direct dependencies between components so they can operate and scale more independently.

Tightly coupled system
Order service directly calls payment service (Example)
so if order service becomes unavailable, payment service fails or wait too.

Decoupled system
Order service places a message in a queue, then payment service processes it when ready

Benefits: It is more reliable, independent for scaling, it has smoother isolation, smoother handling of traffic spikes, asynchronous processing.

Amazon SQS

SQS Definition:
Amazon SQS, Simple Queue Service, is a managed message-queue service used to decouple and connect application components.

SQS uses: Producer: Sends Message | Queue: Stores messages temporarily | Consumer: Receives and processes messages
Producer → Queue → Consumer

Visibility Timeout
Visibility Timeout is the period which received SQS message is temporarily hidden from other consumers.

Dead-letter queue
A dead-letter queue, or DLQ, stores message that could not be processed successfully after repeated attempts.

Standard vs FIFO queues

Standard can be duplicated, it uses best-effort ordering and very high throughput. First in first out is more sequence heavy, supports duplicate prevention because of that.

Standard = speed, FIFO = sequence

Amazon SNS

SNS definition:
Amazon Simple Notification Service, managed publish-subscribe service

SNS fan-out
Fan-out occurs when one published message is delivered to multiple subscribers

Requirement	**Service**
Run event-triggered code without managing servers	AWS Lambda
Provide an HTTP API for Lambda	API Gateway
Temporarily queue work for later processing	Amazon SQS
Preserve strict message order	SQS FIFO
Send one event to multiple subscribers	Amazon SNS
Store repeatedly failing SQS messages	Dead-letter queue
Trigger a function when a file is uploaded	S3 event + Lambda
Broadcast one order event into multiple queues	SNS + SQS

MOCK EXAM

Section A: Precise concepts — 20 marks

Each question is worth 2 marks.

1. Explain the difference between a **public cloud** and **Software as a Service**. [2]

   A public cloud is a deployment method and a Software as a Service, SaaS is a service model.

2. State one clear difference between **scalability** and **elasticity**. [2]
   Scalability is expanding the resources of an application whereas elasticity is increasing or decreasing the available resources to an application based on demand

3. Explain the difference between an AWS **Region** and an **Availability Zone**. [2]
   AWS Region contains multiple Availability Zone in it, an Availability Zone is inside an AWS region and contains one or multiple data centres in it.

4. Explain the difference between an **IAM policy** and an **IAM role**. [2]
   IAM policy is a set of security rules created by the guest for authorization while IAM role is a role to temporarily grant permission access to a certain task.

5. Explain the difference between a relational database's **primary key** and **foreign key**. [2]
   A primary key is a unique key that represents a row in a dataset table. Whereas a foreign key is the key of a primary key representing a foreign table in a table.

6. In DynamoDB, explain the purposes of a **partition key** and a **sort key**. [2]
   The purpose a partition key is to identify keys and a sort key is to search or group partition keys.

7. Explain the difference between **S3 Versioning** and an **S3 Lifecycle rule**. [2]
   S3 Versioning saves the previous version of its data as backup while S3 Lifecycle rule is deleting a data after a certain amount of set time is over or when data gets old.

8. Explain the difference between a **cache miss** and **cache invalidation**. [2]
   A cache miss is when a requested item is missing in the cache and cache invalidation is when user deliberately deletes the cache to retrieve new data from the origin database.

9. In Lambda, explain the difference between an **event** and a **trigger**. [2]
   An event is the description of something that could happen while a trigger is the specific condition that needs to happen to invoke a Lambda Function

10. What is the purpose of the **SQS visibility timeout**, and what happens when it expires before the message is deleted? [2]
    SQS visibility timeout is to allow a consumer to process a message by setting it invisible so other consumers would not process it, if processing fails, it turns back visible to retry and if it expires the message is sent to a dead-letter queue so that it is not process repeatedly and can be investigated later.

Section B: Structured comparisons — 20 marks

Question 11 — EC2 and Lambda [6]

Compare Amazon EC2 and AWS Lambda using three developed differences.
Amazon EC2 is service that supplies virtual machines to customers with its own dedicated resources. AWS Lambda on the other hand is a serverless computing service that is provided to customers. EC2 gives full control to guest's operating system, so guests handle patches and so on. While AWS Lambda handles and manages the server for the user by server provisions, scalability and availability. Amazon EC2 is more suitable for a long-term application that requires persistent server database while AWS Lambda is more suitable for short-term event driven applications like running functions such as notifications, thus should normally remain stateless.

Question 12 — AWS storage [6]

Compare Amazon EBS, Amazon EFS and EC2 Instance Store using:

- Storage type
- Persistence
- Suitable workload

Amazon EBS storage type using a block store type like a hard disk to store its data. EFS uses file storage and EC2 Instance Store is a temporary block store attached to EC2. EBS is persistent, EFS is also persistent but EC2 Instance Store a temporary host block. EBS is suitable for long term storage as it is persistent, EFS is suitable for conveniency and to share a file between multiple separate devices while EC2 Instance Store is suitable for just a temporary host block storage.

Question 13 — RDS features [4]

Compare RDS Multi-AZ and an RDS read replica. State the primary purpose of each and explain how each operates.
RDS Multi-AZ is a RDS feature to have multiple AZ's of a server and RDS read replica is a way to extract read-only queries from a server. By having multiple AZ, it supports availability as another AZ can handle fail over if one AZ fails. RDS real replica reduces load on the RDS as frequently used read-only queries are reduced.

Question 14 — Messaging [4]

Compare Amazon SQS and Amazon SNS using two developed differences.
Amazon SQS is to queue messages to handle one at a time. This decouples computing components and allows more requests than what's naturally capable of the system. SNS is an even announcer where it will announce an event to one or more subscribers. SNS can have multiple SQS as it's subscriber. Amazon SNS is to send out a message to multiple things.

Section C: Case studies — 20 marks

Question 15 — Global online store [10]

An online store has these requirements:

- Web traffic changes unpredictably.
- Traffic must be distributed across several EC2 instances.
- The relational database must survive an Availability Zone failure.
- Product searches are overloading the primary database.
- Product images stored in S3 must load quickly for worldwide users.

Recommend the appropriate AWS service or feature for each requirement and explain how it solves the stated problem.

EC2 Auto Scaling, auto scaling helps to increase and decrease resources based on demand, increasing elasticity, perfect for web traffic that chances unpredictably.
Elastic Load Balancer, Elastic Load Balancer help distribute traffic loads to each EC2 instances to avoid one-sided stress load on one instance.
RDS Multi-AZ, Creates the RDS in multiple Availability Zone so even if one fails, another Availability Zone can handle the fail over.
RDS Read Replica, takes the read queries from the RDS to reduce load by reducing amount of read-only queries.
CloudFront, caches the product images stored in S3 to an edge location nearby to users, allowing for lower latency.

Question 16 — Serverless order processing [10]

A mobile application submits orders through HTTP requests. Sudden traffic must not overload payment processing. Failed payment messages must not retry forever. After an order is completed, email, inventory and analytics systems must all be notified.

Design an architecture using:

- API Gateway
- AWS Lambda
- Amazon SQS
- An SQS dead-letter queue
- Amazon SNS

Explain the complete flow and the role of every service.
First when user performs a request, API Gateway can be the HTTP endpoint, creating it as an API before processing it into a message to be sent into Amazon SQS. Amazon SQS will then decouple the payment messages and other messages by queuing the orders. This allows each computing component to be processed even when there is a sudden surge in order amount, avoiding overloading processes. If a payment process fails, it will not retry forever either as it will be sent to an SQS dead-letter queue for further investigation later. Finally, Amazon SNS can be responsible to announce the event of an order is completed by sending out the notification to the emails, inventory and analytics systems by making them as the subscriber of that SNS.

Mock Review

The partition key determines how DynamoDB distributes data and identifies the partition key-group containing an item. The sort key organizes and distinguishes items sharing the same partition-key value.